SOC Lab Report: Credential Access Detection using Splunk

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1 Abstract

This report demonstrates detection of credential access attempts in a SOC environment using Splunk. Attackers often attempt to access sensitive processes like LSASS to extract credentials. This scenario simulates such behavior and analyzes logs to detect suspicious activity.

2 Objective

To detect credential access attempts by monitoring process activity targeting LSASS using Splunk.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Credential Access
- Technique: OS Credential Dumping (T1003)

5 Attack Simulation

Credential access behavior was simulated by querying LSASS process information.

5.1 Commands Used

```
tasklist | findstr lsass
powershell -Command "Get-Process lsass"
```

5.2 Explanation

These commands enumerate the LSASS process, which is commonly targeted by attackers to extract credentials.

5.3 Screenshot

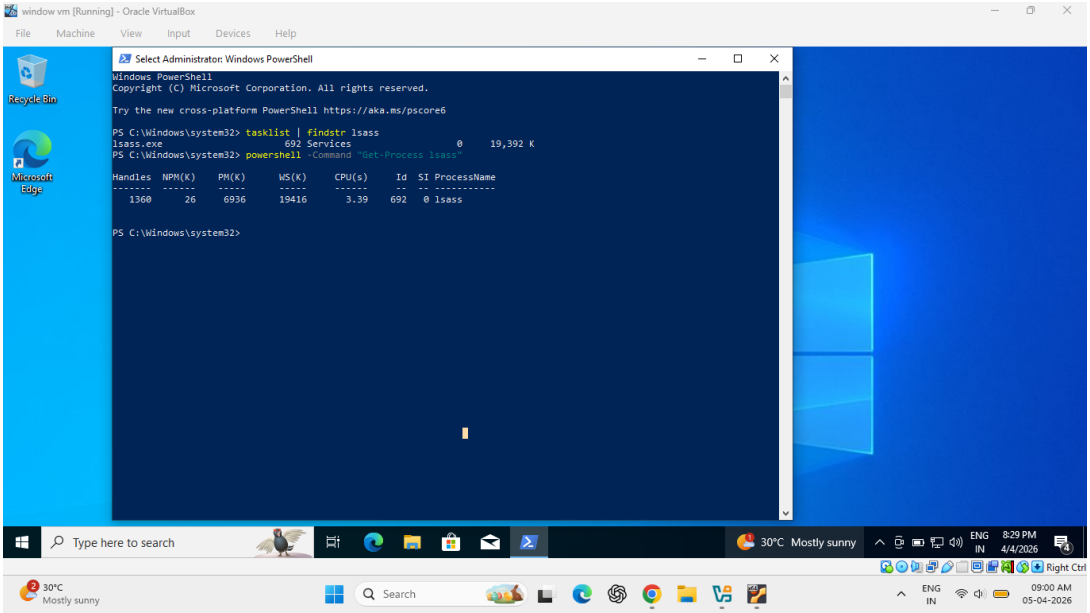

Figure 1: LSASS Process Enumeration

6 Detection in Splunk

Credential access attempts are detected using process creation logs (EventCode 4688).

6.1 Detection Query

```
index=windows EventCode=4688
| search Command_Line="*lsass*" OR Command_Line="*tasklist*" OR Command_Lin
| table _time, user, Command_Line
```

6.2 Explanation

The query identifies suspicious commands that interact with the LSASS process.

6.3 Screenshot

7 Investigation

Further investigation was performed to analyze suspicious activity.

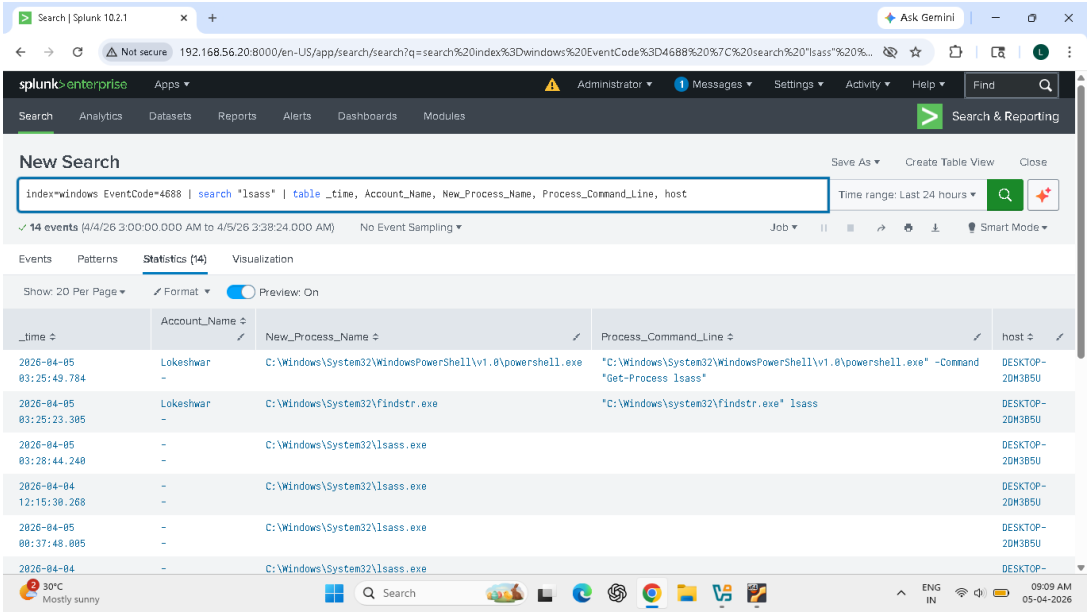

Figure 2: Detection of Credential Access Activity

7.1 Timeline Query

```
index=windows user="TARGET_USER"
| sort _time
| table _time, EventCode, Command_Line
```

7.2 Findings

- Commands targeting LSASS process detected
- Possible credential enumeration activity
- Suspicious behavior observed

7.3 Screenshot

7.4 Screenshot

8 Response Actions

Immediate actions were taken to mitigate the threat.

- Terminated suspicious processes
- Reset user credentials
- Enabled monitoring

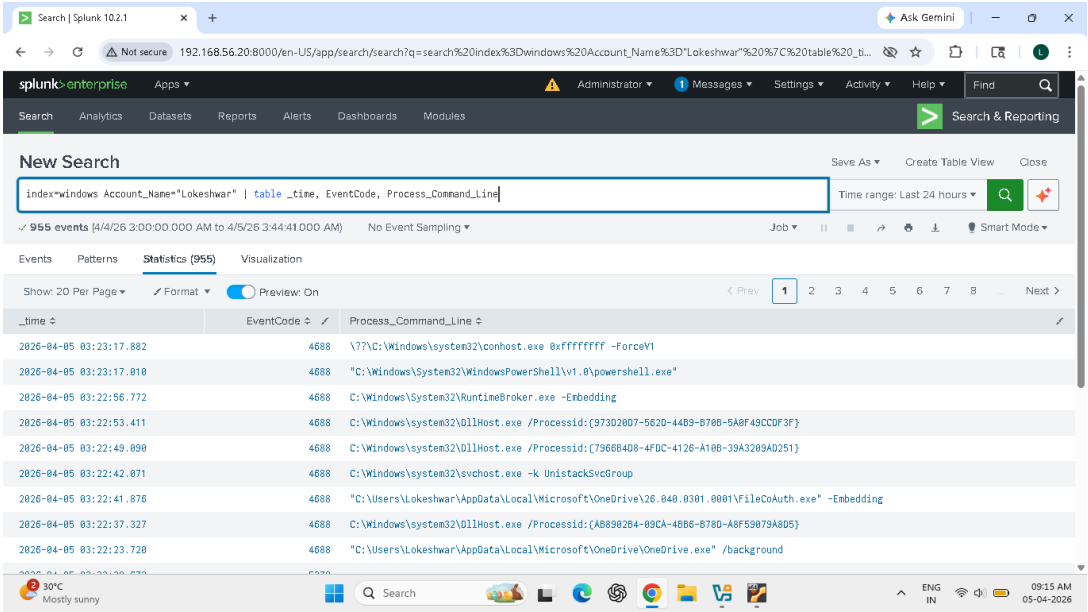

Figure 3: Timeline showing LSASS-related activity

8.1 Commands Used

```
taskkill /IM powershell.exe /F
net user <username> NewPassword@123
```

8.2 Explanation

The suspicious process was terminated to prevent further credential access attempts. User credentials were reset to secure accounts.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

Suspicious activity targeting the LSASS process was detected, indicating a potential credential access attempt.

9.2 Impact

User credentials could be compromised if such activity is not mitigated.

9.3 Conclusion

The activity was successfully detected and mitigated using Splunk SIEM.

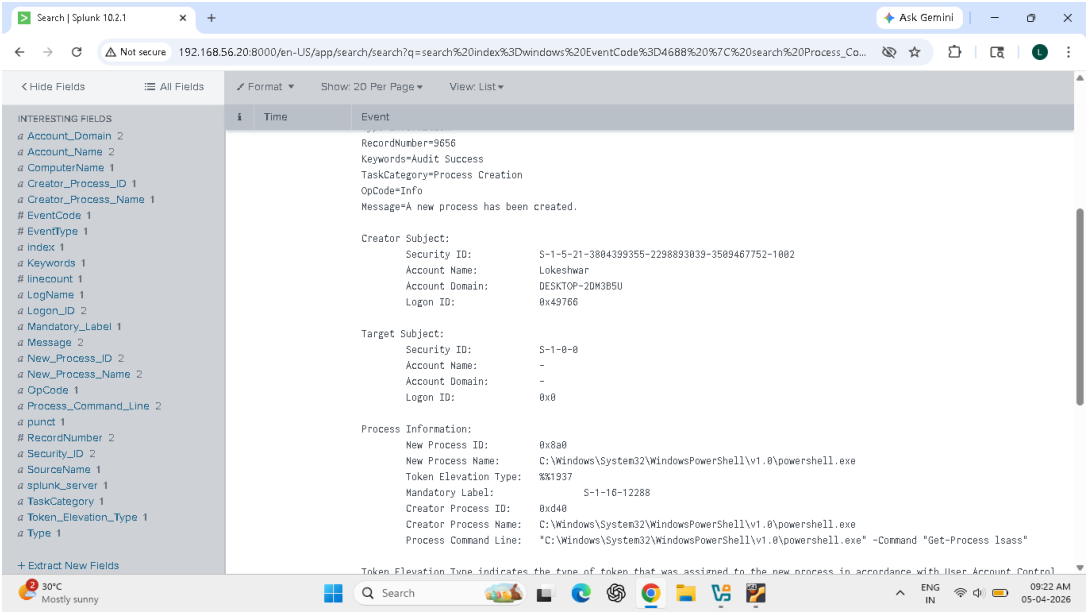

Figure 4: Timeline showing LSASS-related activity

10 Conclusion

This scenario highlights the importance of monitoring sensitive processes like LSASS to detect credential access attempts.

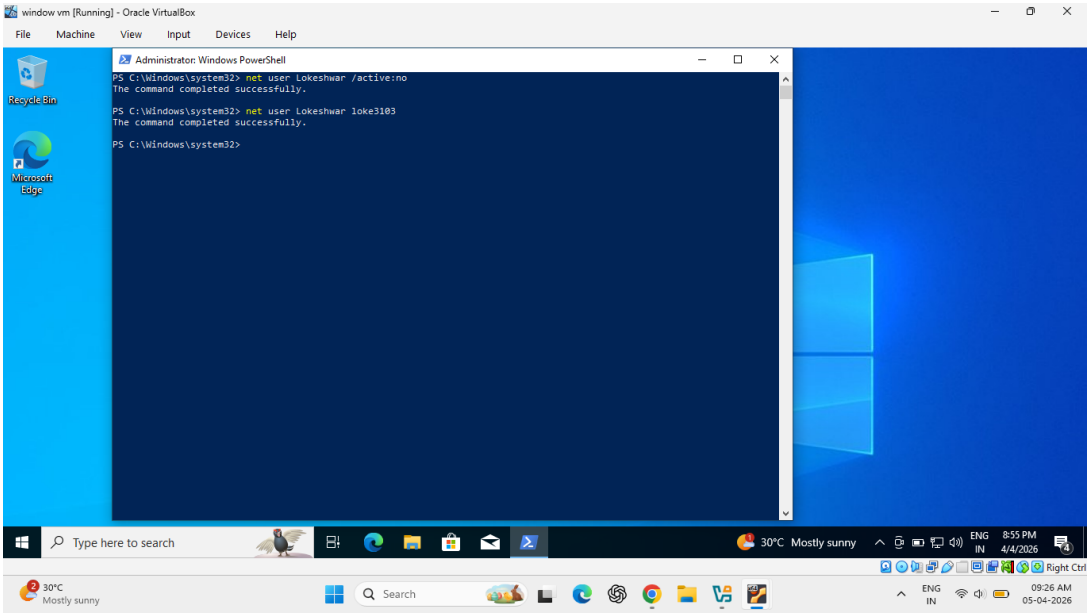

Figure 5: Response Actions - Process Termination and Credential Reset